SUR 2025/2026 Project

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1. Augmentation

The training pool is small (30 base paired stems of *m431* across two sessions, ~190 non-target stems from disjoint speakers) and the assignment warns the eval set will contain input-quality variations not seen at training. Offline augmentation (`scripts/augment_dataset.py`) addresses both: it enlarges the effective training set and regularises against acoustic and visual distortions. Audio augmentation applies time stretch (0.9–1.1×), gain (±6 dB), polarity flip ($p=0.5$), peaking-band equalisation (EQ; ±3 dB at 200–4000 Hz) and additive White Gaussian Noise (WGN) (SNR 15–30 dB). image augmentation applies horizontal flip ($p=0.5$), rotation (±10°), translation (±5 %), brightness and contrast scaling (0.85–1.15×) and per-pixel WGN ($\sigma \in [0,0.04]$). Each base target sample receives 15 augmented variants and each non-target receives 5.

2. Evaluation Metrics

Four quantities are reported throughout. **AUC** is the primary rank statistic, threshold-free and robust to class imbalance. Most variant-selection decisions (audio feature subsets, image regularisation, GMM-UBM mixture counts) were made on AUC. **Equal Error Rate (EER) and the EER threshold** summarise the symmetric-cost operating point under prior 0.5. Drift of the EER threshold away from zero flags miscalibration, and we rejected the MAP-adapted GMM-UBM variant on this signal alone (its EER threshold sat at $+0.25$ across the entire τ sweep). **Confusion at *logit = 0*** (TP/FP/FN/TN) gives the operationally relevant quantities under the assignment's hard rule. **Precision and F_1** become decisive when stress-dev separates the fusion variants: `fusion_3way` reaches precision 0.70 and $F_1 = 0.70$ on stress dev versus 0.31 / 0.47 for the more recall-favouring fusions, which is what motivates keeping it in the submitted lineup despite a slightly lower stress AUC than `fusion_hybrid_image`.

3. Image System

Image features are 9-bin Histograms of Oriented Gradients over 8×8-pixel cells with 2×2-cell L2-Hys normalised blocks on 80×80 grayscale crops, producing 2916-d vectors. A balanced logistic regression with L2 regularisation ($C=0.1$) classifies them. Hard decisions threshold *decision_function* at zero. The progression from the syllabus baseline isolates three design changes:

Step	Description	AUC	EER
1	Baseline: pixels → PCA(D=21) → Gaussian LDA + 2 IRLS	0.943	20.00 %
2	HOG → PCA(D=21) → Gaussian LDA + 2 IRLS	0.977	9.17 %
3	HOG (full 2916-d) → balanced LogReg, $C=0.1$, no augmentation	0.903	21.67 %
4	+ offline augmentation (submitted)	1.000	0.00 %

Step 1→2 isolates the HOG vs. raw-pixel feature gain at fixed dimensionality. Step 2→3 is a *regression*: dropping PCA in favour of full-dimensional HOG with L2 leaves the threshold-zero hard rule mis-calibrated at only 10 target training images. Step 3→4 shows that augmentation is what makes the high-dimensional architecture viable. Full-dimensional + L2 in isolation is worse than the PCA-bridged baseline classifier of step 2.

4. Audio System

Both audio systems share the same frontend: an energy + zero-crossing-rate voice activity detector (VAD), 13 mel-frequency cepstral coefficients at 25 ms / 10 ms framing with per-sample cepstral mean–variance normalisation, and probabilistic-YIN F_0 estimation over voiced frames in 60–400 Hz. From the F_0 track and frame energies we compute five prosodic statistics (*log F_0* mean and standard deviation, jitter, shimmer, voicing ratio).

Path A, aggregated features and logistic regression. The 5 prosodic statistics and the 13 MFCC means form an 18-d sample-level vector classified by balanced LogReg (*C=1.0*). This is shipped as `audio_logreg.txt`.

Path B, Gaussian Mixture Model with universal background, MAP-free. Following we fit a 16-mixture diagonal-covariance universal background model (UBM) on non-target frames and a 4-mixture diagonal-covariance target GMM on *m431* frames, *independently* by EM. We chose MAP-free over MAP adaptation because ~32 k augmented target frames are plentiful at the per-component level ($\alpha_k \to 1$ regardless of relevance factor), and a relevance-factor sweep over $\tau \in \{5, 10, 20, 100, 500, 2000, 10000\}$ produced an AUC plateau of 0.98 with the threshold-zero hard rule miscalibrated (EER threshold drifted to *+0.25*). The per-sample score is the frame-averaged log-likelihood ratio. We did not attempt i-vector front-end factor analysis, the standard upgrade over GMM-UBM, because it requires hundreds of background speakers for the total-variability matrix to be estimable, whereas our pool has 19 distinct non-target speakers.

The VAD-empty pathology. Auditing eval revealed 79 of 1346 samples (5.9 %) producing identical audio score ≈ *+1.535*. These extract to all-zero feature vectors: under heavy gain attenuation or short voiced segments, the energy-relative VAD trims every frame, leaving prosodic and MFCC statistics at their zero-fallback. The trained classifier maps this zero input to a bias-driven default that happened to be positive, mislabelling all 79 as target. Spectral-flatness and voiced-fraction analysis confirmed all 79 are real speech, not white noise (median voiced fraction 0.60). The sentinel default was discarding genuine information. The same pathology contaminates training: 121 of 1112 augmented training samples also extract to zero vectors, with the target class disproportionately affected (81 zero-frame target augmentations vs. 40 non-target), so the model also *learned* a small "zero ⇒ target" correlation as a class-asymmetric augmentation artefact.

Hybrid VAD-on / VAD-off fallback. The fix is a routing scheme (`scripts/audio_gmm_hybrid.py`): an auxiliary 16+4 GMM-UBM is trained on the same MFCC frames *without* applying VAD. At scoring time the submitted VAD-on model is consulted first, and the no-VAD model takes over only when VAD has trimmed every frame. The hybrid is shipped as `audio_gmm_hybrid.txt` and short-circuits both the eval-time sentinel defaults and the train-time spurious correlation at scoring.

The audio progression covers both clean and stress development splits:

Step	Description	Clean AUC	Stress AUC
1	Baseline: course MFCC + 2 GMMs M=30 (no VAD, no CMVN)	0.978	0.799
2	+ frame-energy VAD + per-sample CMVN, same 2 GMMs M=30	1.000	0.730
3	Submitted GMM-UBM (MAP-free, sentinel for VAD-empty)	0.997	0.463
4	Hybrid VAD-on/off (submitted)	0.997	**0.870**

VAD and CMVN saturate clean dev (steps $1 \to 2$) but degrade the stress split because they introduce the failure mode that step 4 must repair. The hybrid recovers the baseline's stress robustness without sacrificing clean-dev performance.

5. Multimodal Fusion

Fusion is performed at the score level. For each per-modality system, dev-set scores are z-normalised. A balanced logistic regression is fitted on the concatenated z-scores and the ground-truth labels. The classifier is then applied to z-normalised eval scores. Score-level was preferred to feature-level because the audio (18-d) and image (2916-d) spaces are extremely imbalanced: stacking them with ~1100 training samples invites the curse of dimensionality, whereas a 2- or 3-feature fusion logreg has too few parameters to overfit the 70-sample dev split. We submit three variants: `fusion_logreg.txt` over $(z(\text{audio_logreg}), z(\text{image}))$. `fusion_hybrid_image.txt` over *(z(audio_gmm_hybrid), z(image))*. And `fusion_3way.txt` over $(z(\text{audio_logreg}), z(\text{audio_gmm_hybrid}), z(\text{image}))$. The three-way variant tests whether the two audio paths capture complementary signal, since aggregated 18-d features and frame-level GMM likelihoods are structurally different objects.

6. Validation Strategy

Clean `*_dev` (n=70) saturates for image and fusion, so variant selection ran on a *stress* dev built by re-augmenting dev with a *disjoint* envelope (gain *-15* to *-8* dB, SNR 3–13 dB, time-stretch 0.78–0.88, image brightness 0.45–0.70, per-pixel noise $\sigma \in [0, 0.10]$), which converts each sample from interpolation to extrapolation under the trained model. Each variant in the per-modality sweeps was promoted into the submitted lineup on its stress AUC, with stress EER and threshold-zero confusion as tiebreakers. The 50/50-prior decision rule is enforced at training (`class_weight='balanced'`) and inference (`hard_threshold_logit = 0.0`).

7. Results

The submission ships one image system, two audio systems and three fusion variants, 6 result files in total.

Based on our testing, all three fusion variants saturate clean dev. The image system is essentially perfect on clean conditions, the GMM-UBM-hybrid audio system is close behind, and the LogReg audio system trails by a few AUC points. Stress dev separates the variants clearly, and the spread there is the better generalisation indicator than clean-dev AUC. Under stress, the hybrid audio path strongly outperforms the LogReg audio path, and the fusion variants that use it inherit that advantage. We expect `fusion_hybrid_image` to maximise stress AUC and `fusion_3way` to dominate on precision-weighted metrics: `fusion_3way` produces a substantially more balanced confusion matrix on stress, trading a small AUC decrement for a large precision gain. Under an operating point that penalises false positives more than the assignment's symmetric costs, `fusion_3way` would be the strongest single submission.

8. Reproduction

```
uv run scripts/augment_dataset.py --config configs/augment_audio.toml
uv run scripts/augment_dataset.py --config configs/augment_image.toml
uv run scripts/train.py --config configs/audio.toml
uv run scripts/train.py --config configs/image.toml
uv run scripts/train.py --config configs/fusion.toml
uv run scripts/evaluate.py --config configs/audio.toml --split eval
uv run scripts/evaluate.py --config configs/image.toml --split eval
uv run scripts/evaluate.py --config configs/fusion.toml --split eval
uv run scripts/audio_gmm_pipeline.py --mode map_free
uv run scripts/audio_gmm_hybrid.py
uv run scripts/fix_vad_empty.py
uv run scripts/fusion_hybrid_image.py
uv run scripts/fusion_3way.py
```

All randomness is seeded. Reproduction is deterministic from a fixed seed.